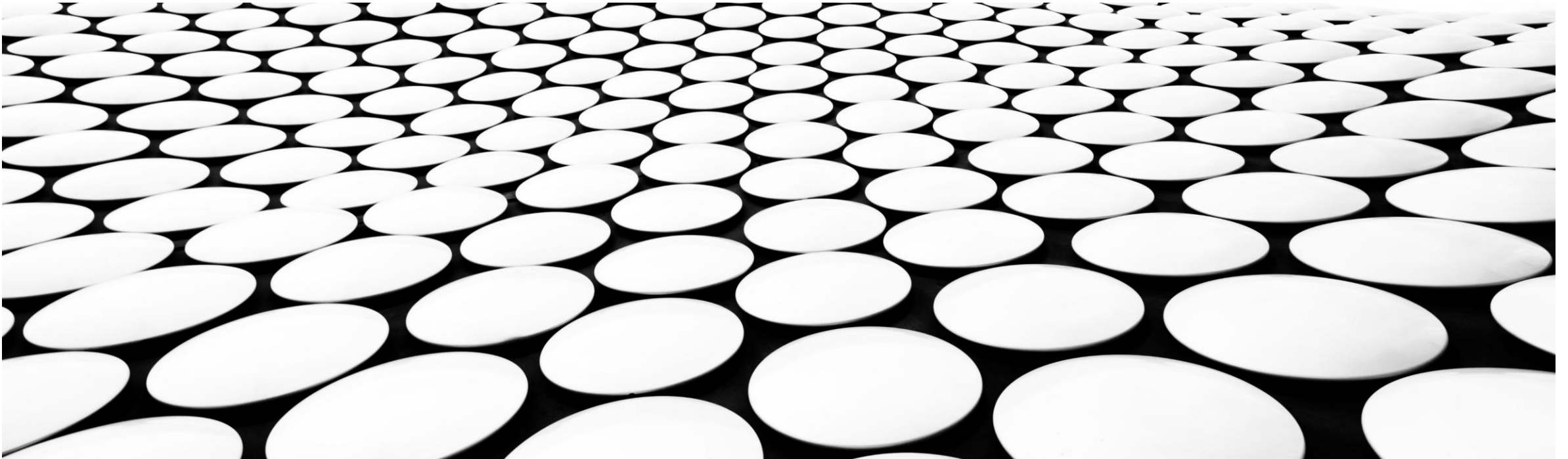
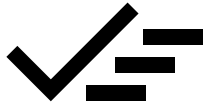


PREDICTIVE ANALYSIS

A focus on Supervised Machine Learning





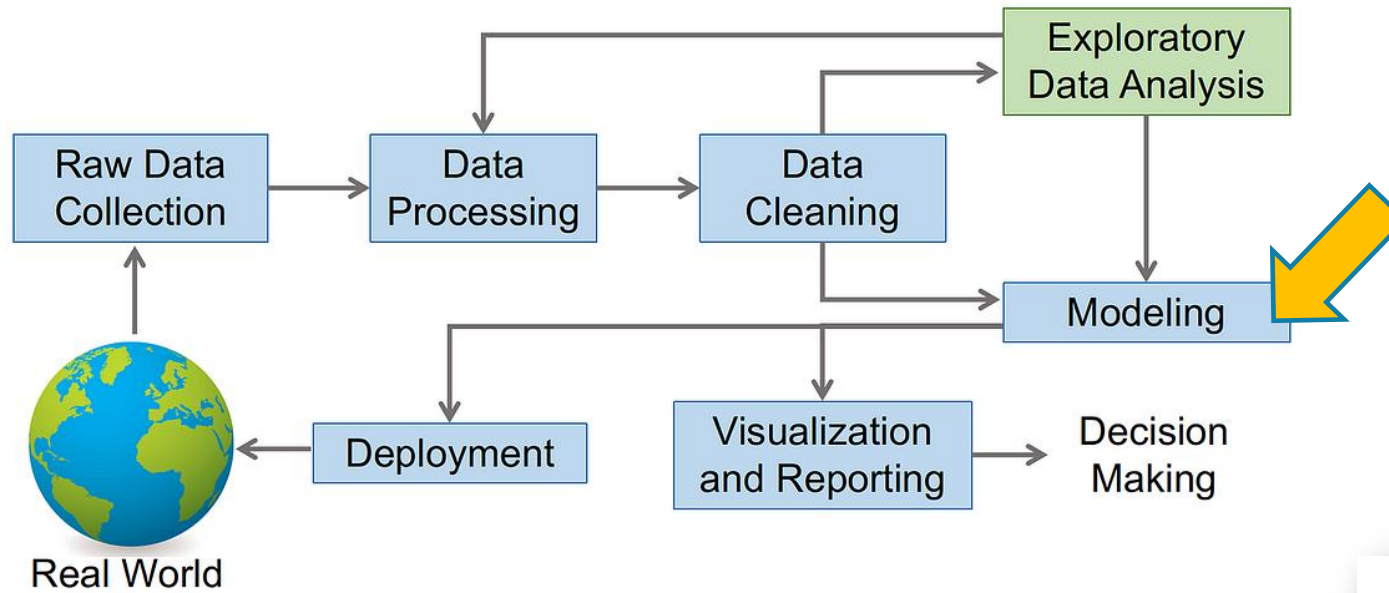
PREDICTIVE ANALYSIS

- Introduction to Machine Learning
- Supervised Machine Learning steps

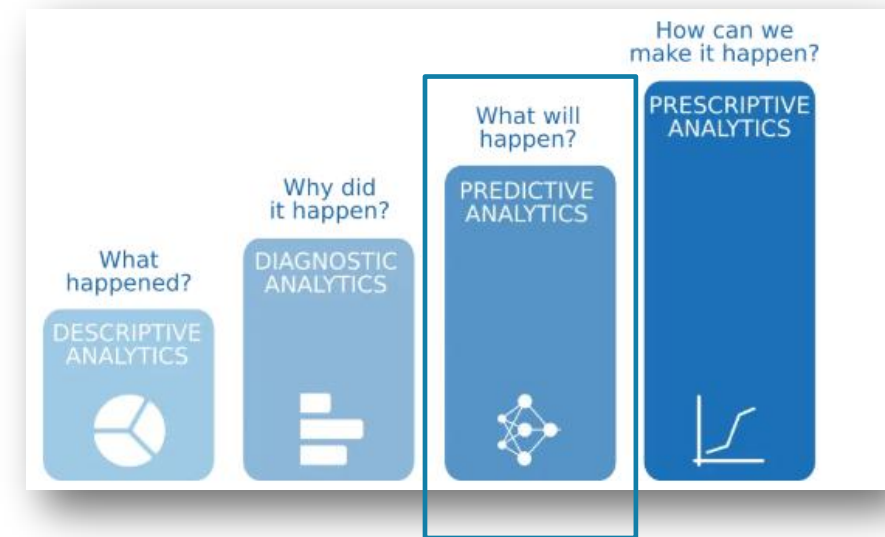
Predictive analysis

Introduction to Machine Learning

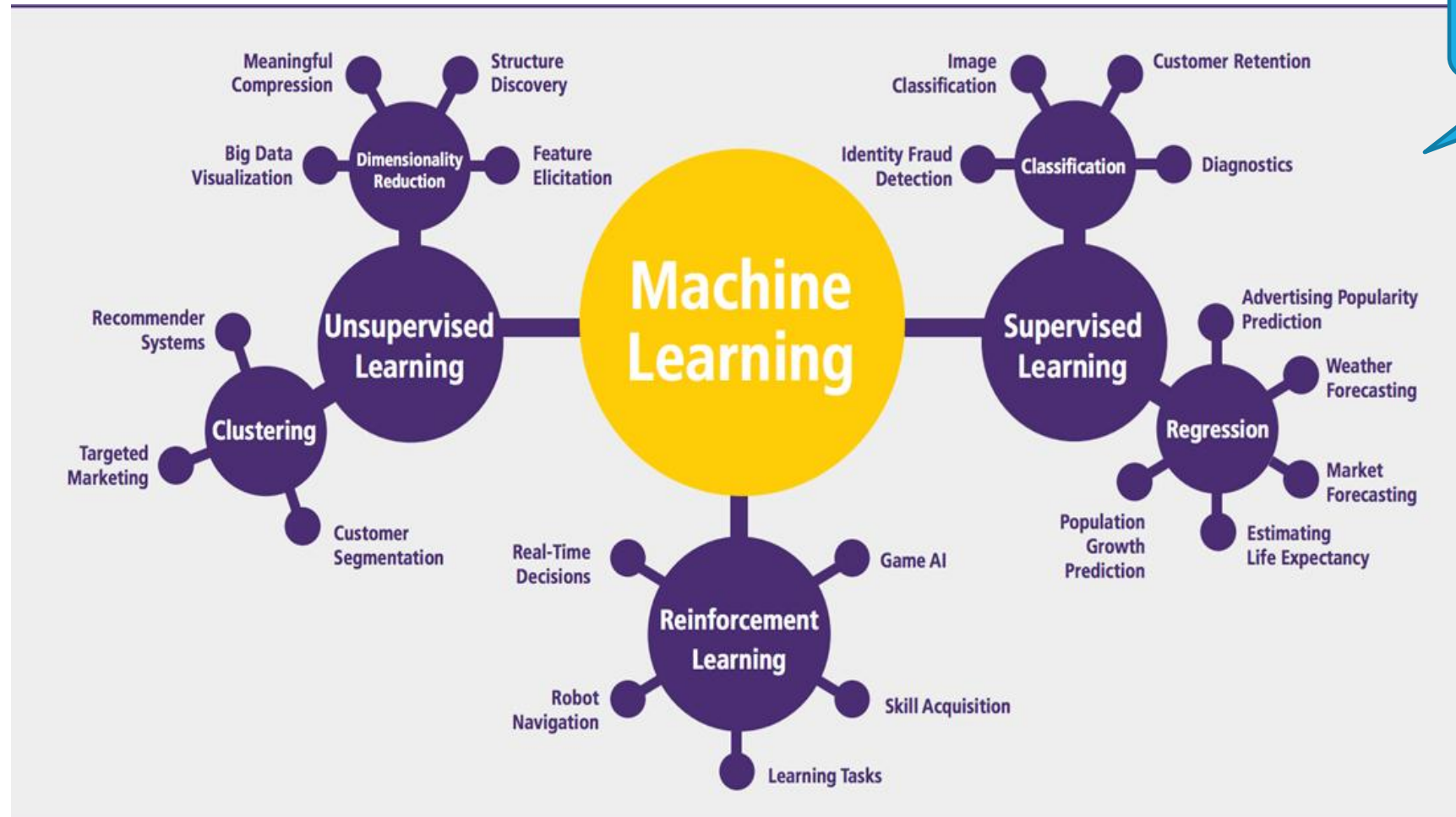
Data Science Process

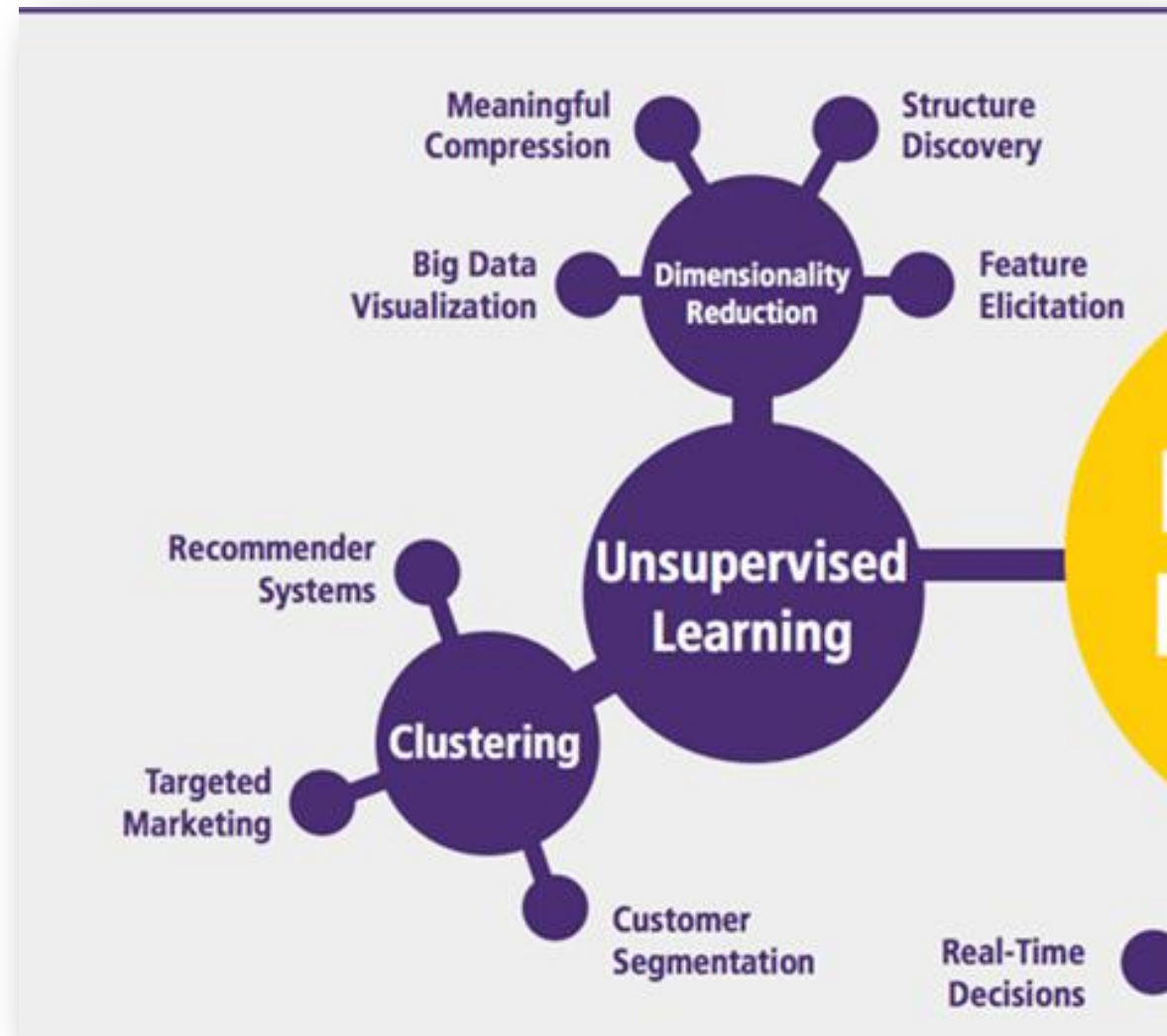


Associated with
Machine Learning



Different types of ML





Machine Learning

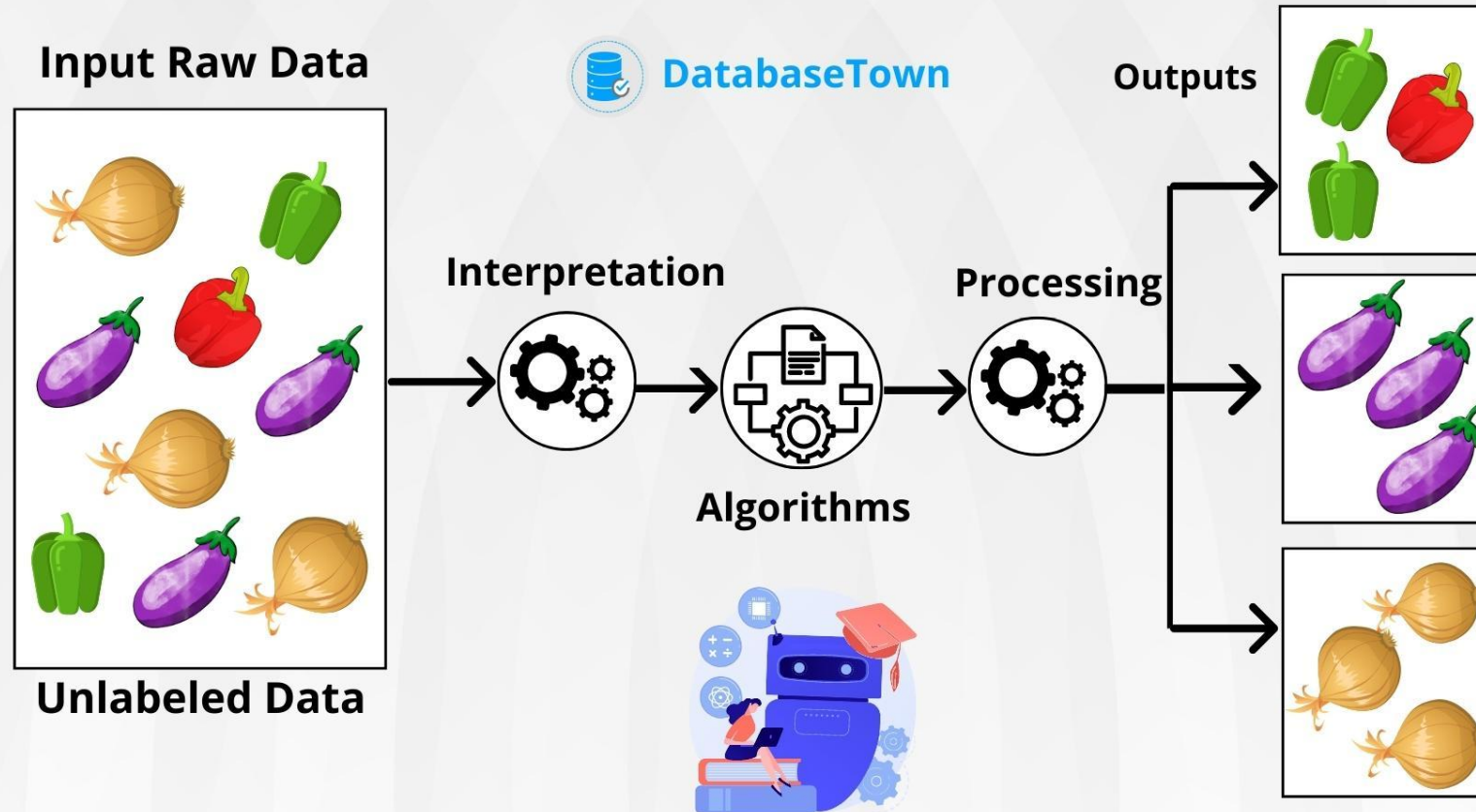
Unsupervised Learning: Types, Applications & Advantages

admin May 27, 2023

[Source](#)

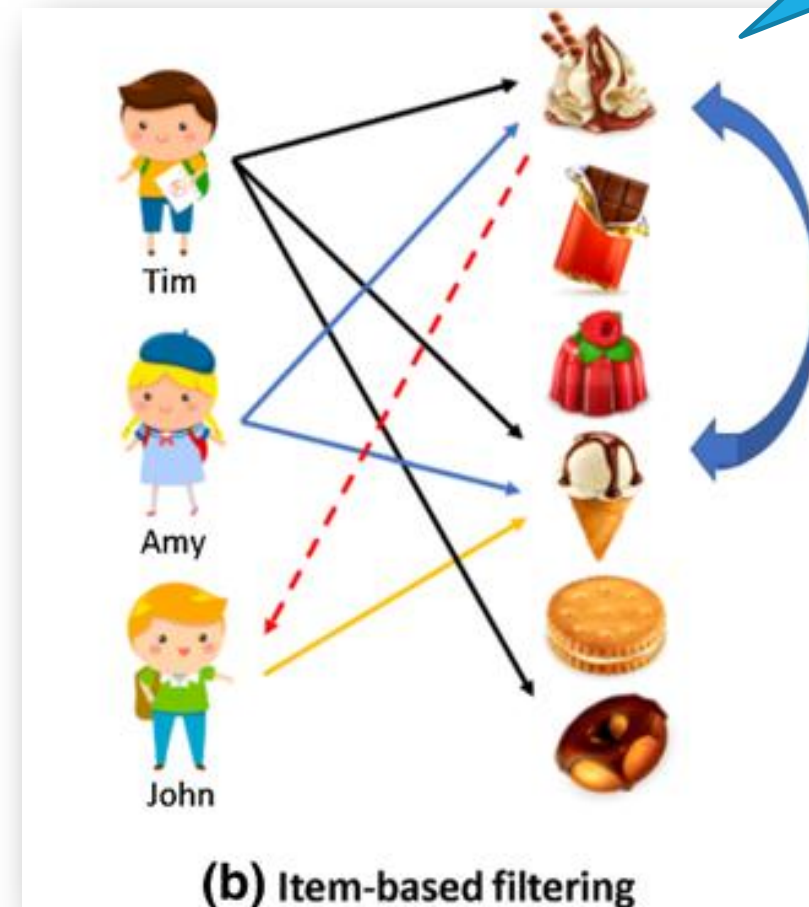
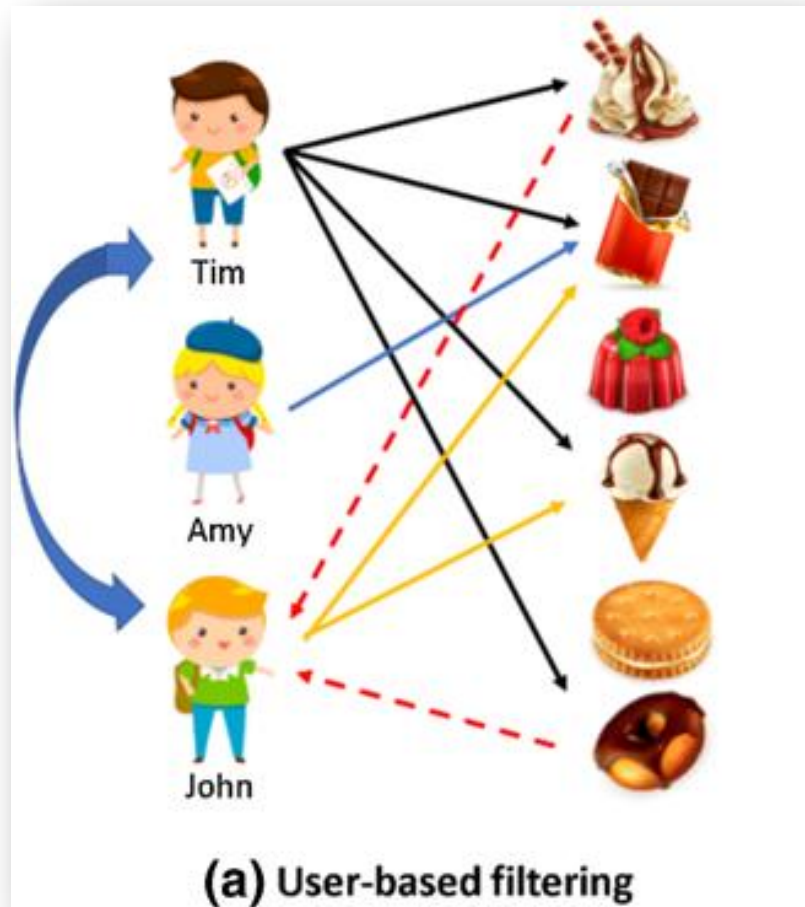
UNSUPERVISED LEARNING

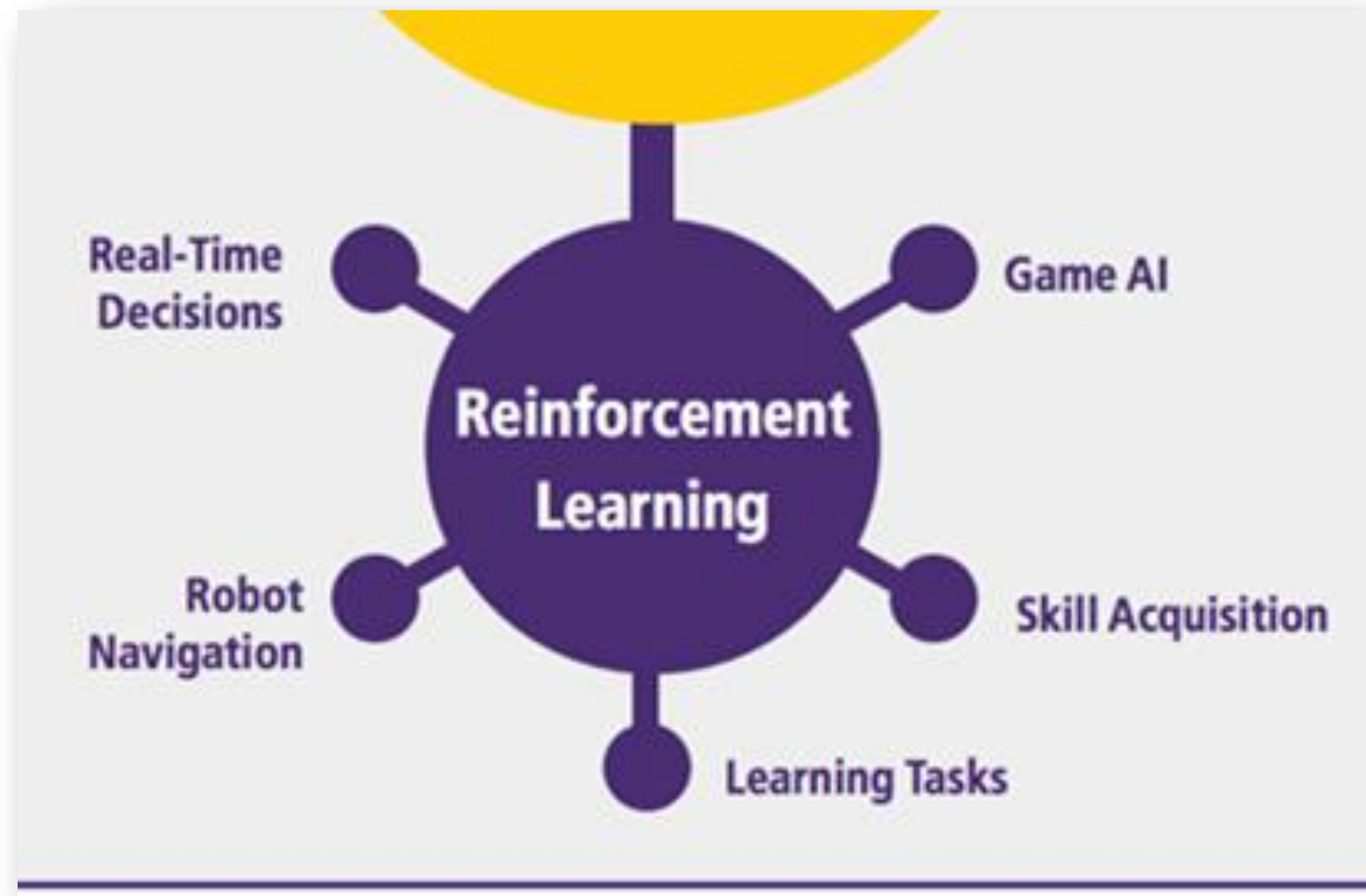
Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.

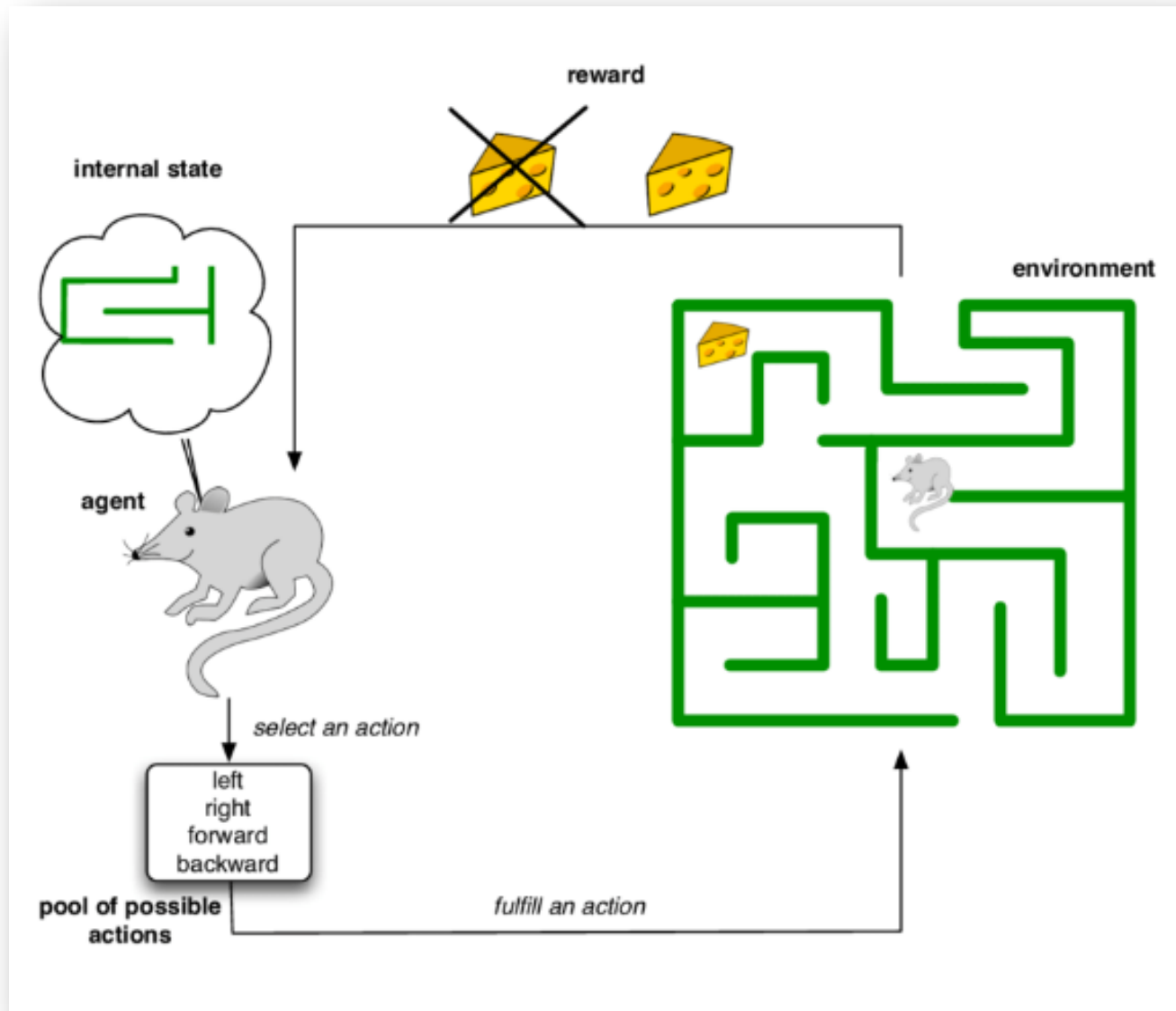


Clustering

Recommender systems

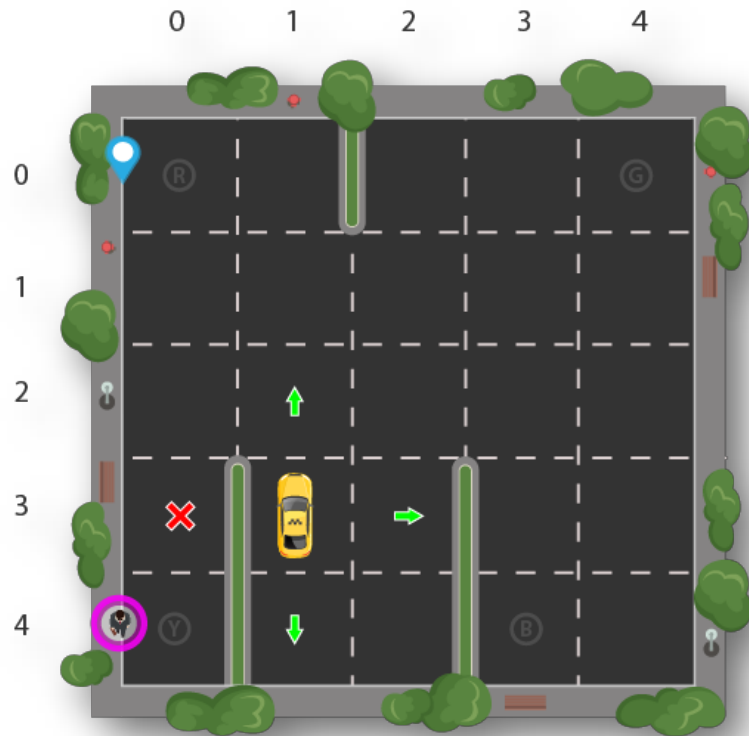




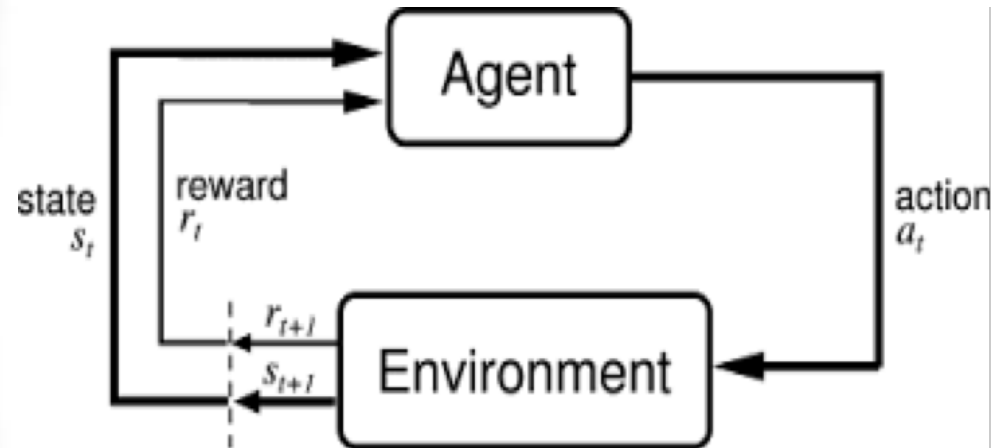


Scenario with a long
term goal

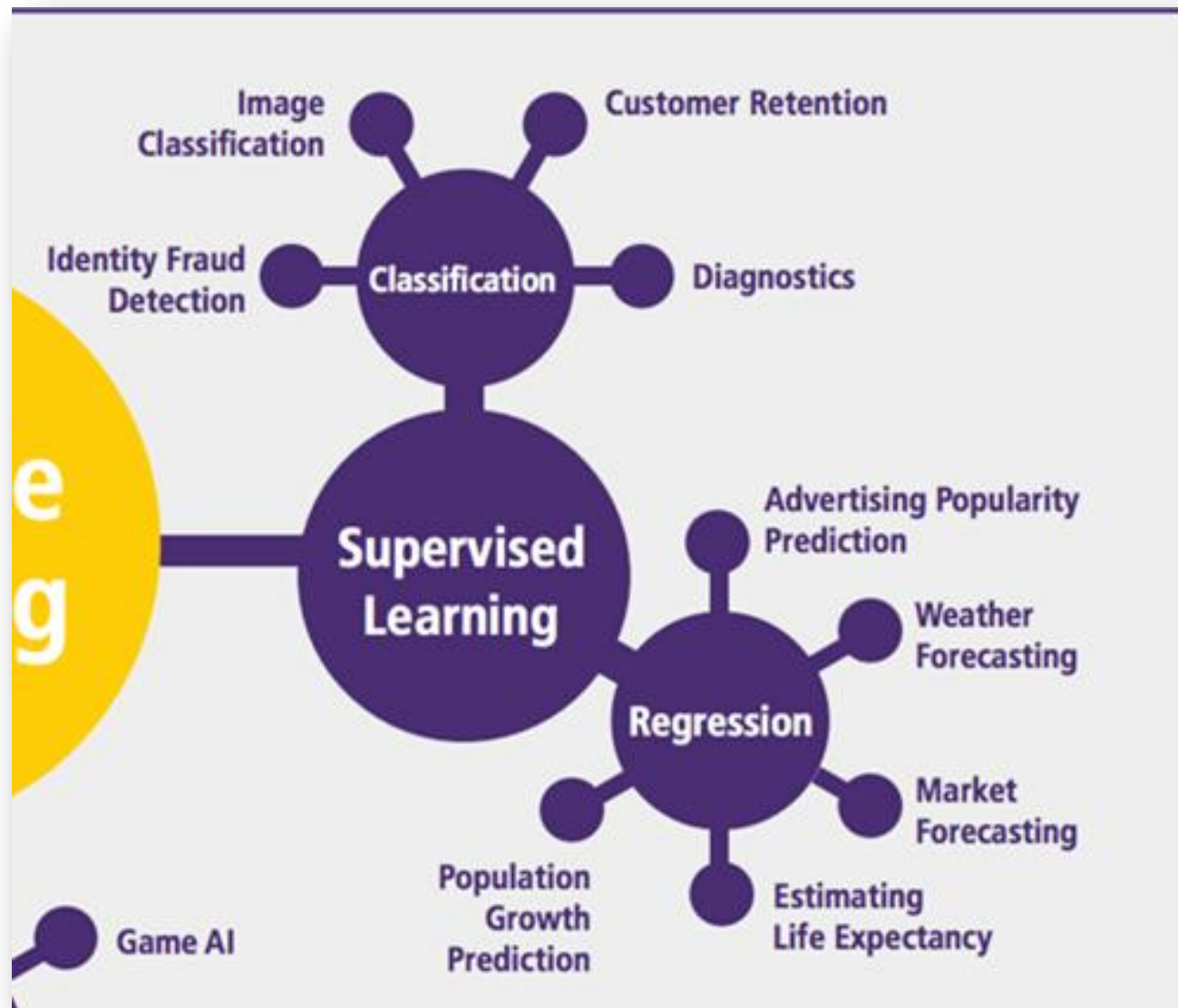
Self-driving cars



Typical application



Source



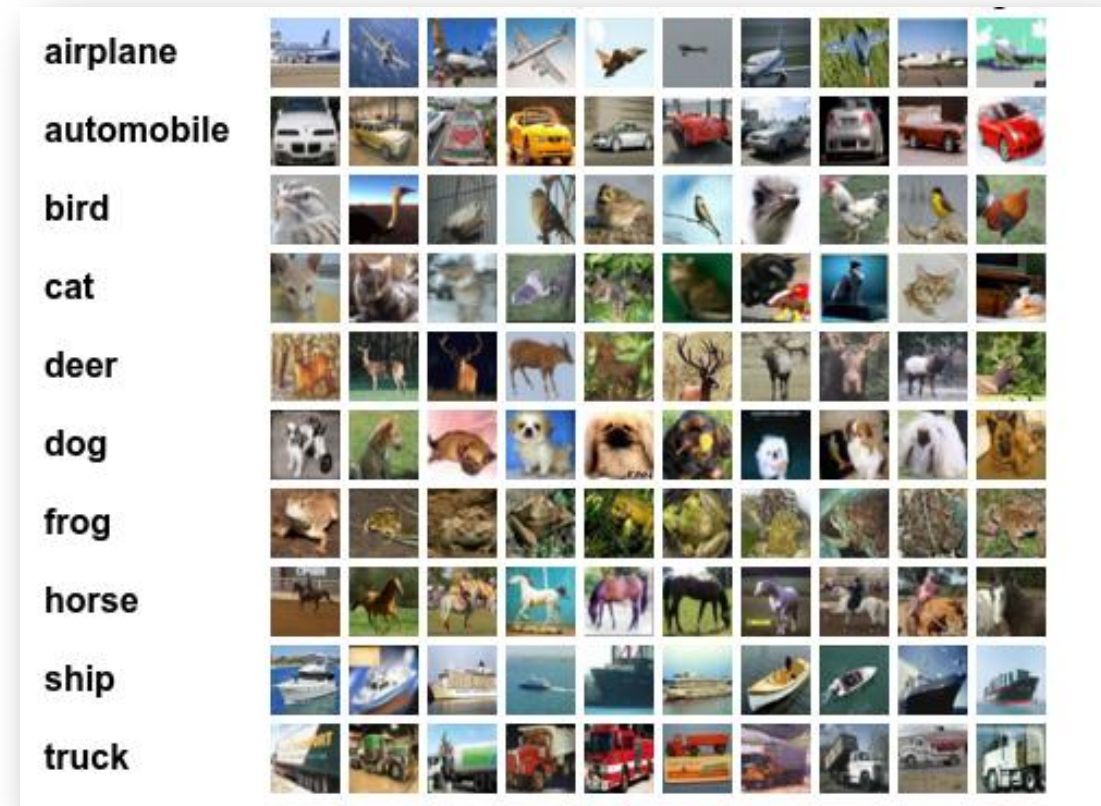
Classification: the target is the index of a Class

Example: Image Recognition

x: grid (32x32) of colored pixel values

t: 10 classes

$$t \in 1, \dots, K$$



[Source](#)

Regression: the target is a real number

1. Medical Insurance Cost Prediction Dataset

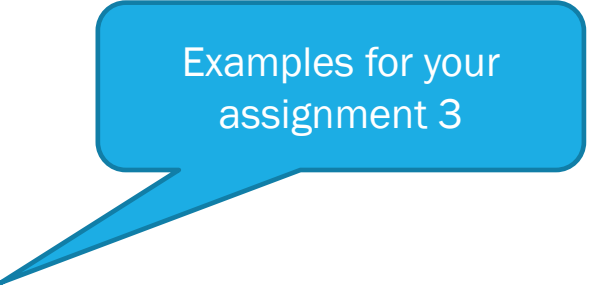
- **Description:** Predicts medical insurance costs based on age, BMI, smoking status, region, and other features.

2. Car Price Prediction Dataset

- **Description:** Predicts car prices based on features like mileage, brand, year, fuel type, and transmission type.

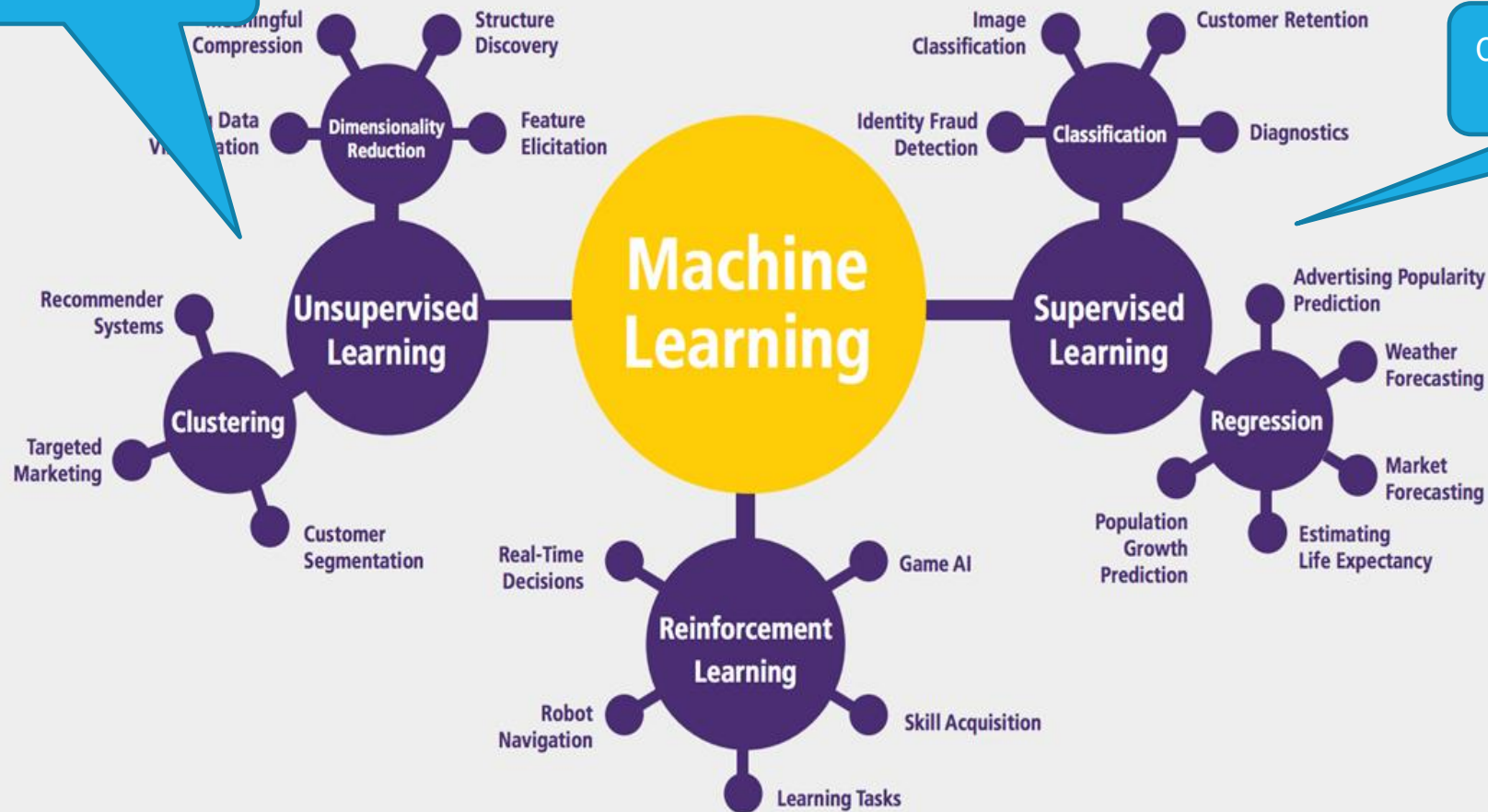
3. Housing Prices Regression Dataset

- **Description:** Predicts house prices with a couple of binary features.



Examples for your
assignment 3

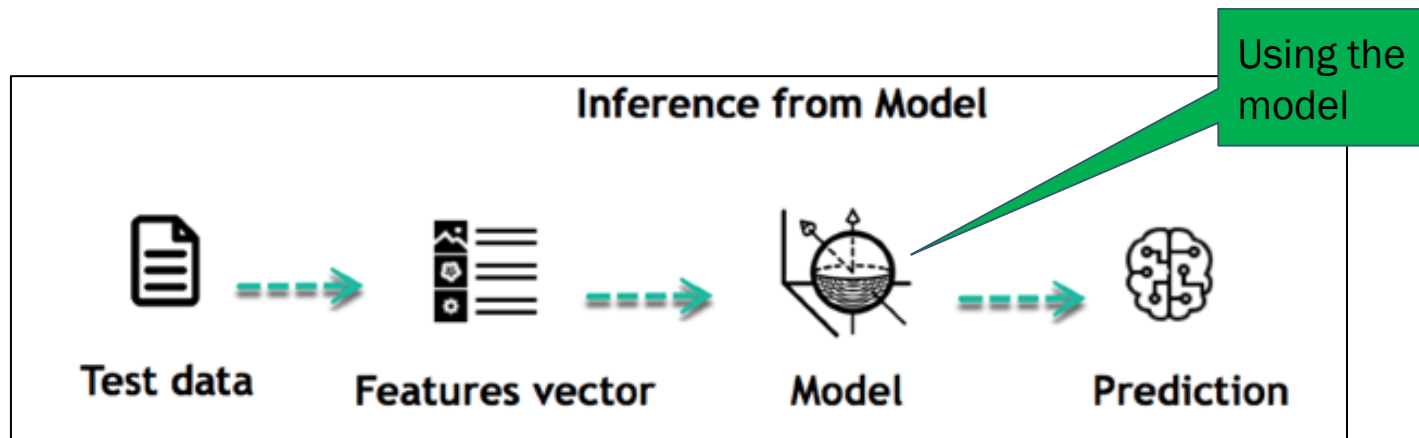
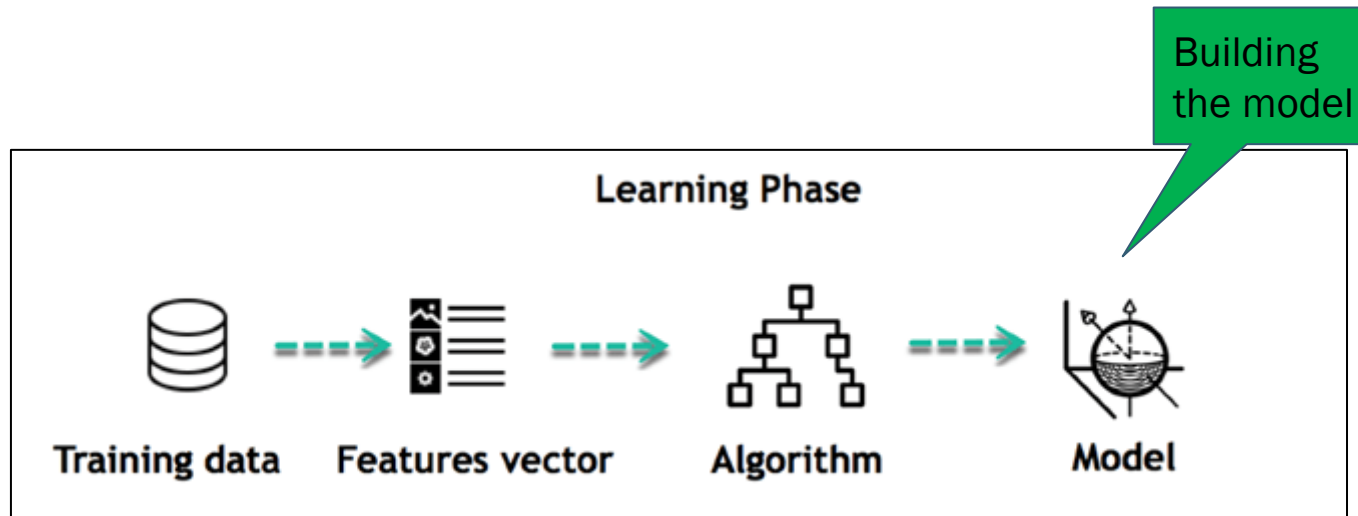
Covered week of March 18th

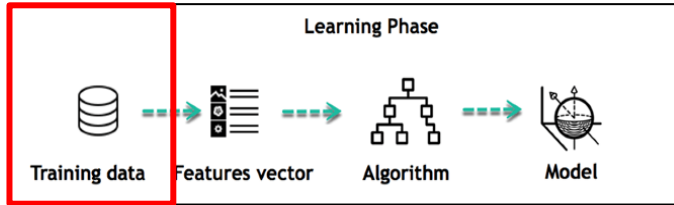


Covered weeks of March 4th and March 11th

Supervised ML

Empirical Framework



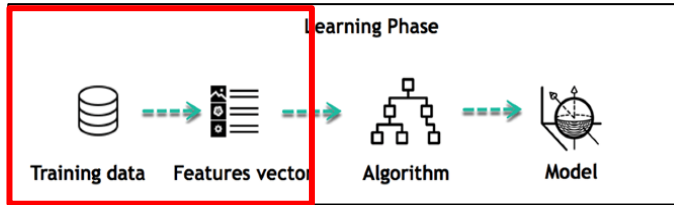


Gathering of annotated data

	Possible output	
Movie	<u>Classification</u> Genre	<u>Regression</u> Gross Revenue
M1	Drama	3.2M
M2	Drama	200K
M3	Comedy	15M
M4	Drama	500K
M5	Drama	10K
M6	Comedy	1.3M
M7	Comedy	870K

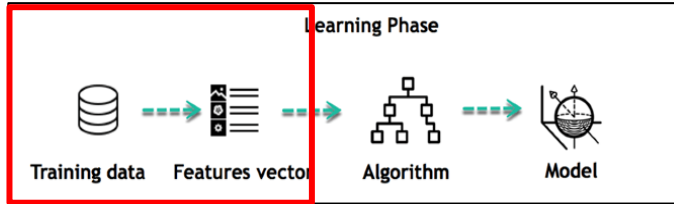
	Possible classes	
Movie	Genre	Viewer's appreciation
M1	Drama	Bad
M2	Drama	Bad
M3	Comedy	Bad
M4	Drama	Bad
M5	Drama	Bad
M6	Comedy	Bad
M7	Comedy	Good

Unbalanced training set



Representation of the data

	Features			Possible output	
Movie	Duration	Producer	Lead actor/actress	<u>Classification</u> Genre	<u>Regression</u> Gross Revenue
M1	90	P1	A1	Drama	3.2M
M2	140	P3	A2	Drama	200K
M3	80	P1	A1	Comedy	15M
M4	90	P1	A3	Drama	500K
M5	100	P3	A4	Drama	10K
M6	110	P2	A5	Comedy	1.3M
M7	100	P2	A5	Comedy	870K

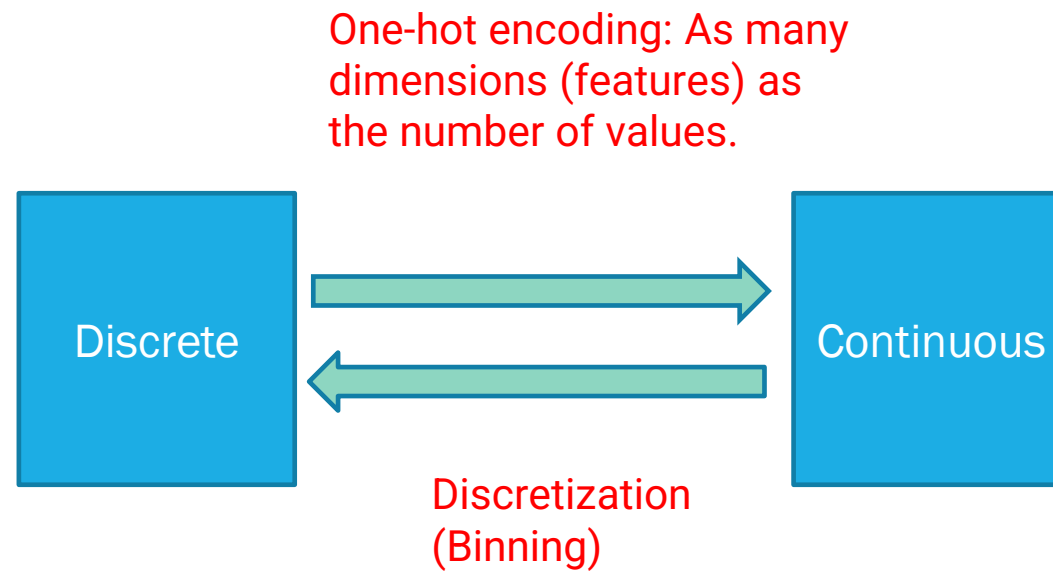


	Features			Possible output	
Movie	Duration	Producer	Lead actor/actress	<u>Classification</u> Genre	<u>Regression</u> Gross Revenue
M1	90	P1	A1	Drama	3.2M
M2	140	P3	A2	Drama	200K
M3	80	P1	A1	Comedy	15M
M4	90	P1	A3	Drama	500K
M5	100	P3	A4	Drama	10K
M6	110	P2	A5	Comedy	1.3M
M7	100	P2	A5	Comedy	870K

Continuous
features

Discrete
features

Feature transformation



	Features		
Movie	Duration	Producer	Lead actor/actress
M1	90	P1	A1
M2	140	P3	A2
M3	80	P1	A1
M4	90	P1	A3
M5	100	P3	A4
M6	110	P2	A5
M7	100	P2	A5

ONE-HOT ENCODING

Movie	Length	P1	P2	P3	A1	A2	A3	A4	A5	Class
M1	90	1	0	0	1	0	0	0	0	Drama
M2	140	0	0	1	0	1	0	0	0	Drama
M3	80	1	0	0	1	0	0	0	0	Comedy
M4	90	1	0	0	0	0	1	0	0	Drama
M5	100	0	1	0	0	0	0	1	0	Drama
M6	110	0	1	0	0	0	0	0	1	Comedy
M7	100	0	1	0	0	0	0	0	1	Comedy

Producer and actor features transformed in one-hot vectors.

Movie	Duration	Discretized Duration	Producer	Lead actor	TARGET
M1	90	B1	P1	A1	Drama
M2	140	B3	P3	A2	Drama
M3	80	B1	P1	A1	Comedy
M4	90	B1	P1	A3	Drama
M5	100	B2	P2	A4	Drama
M6	110	B2	P2	A5	Comedy
M7	100	B2	P2	A5	Comedy

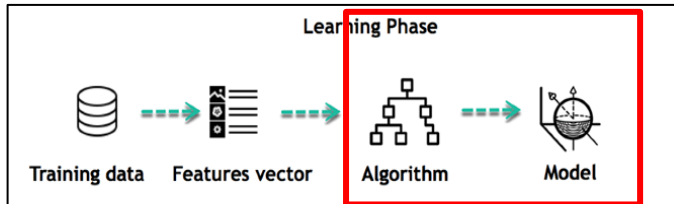
DISCRETIZATION / BINNING

$B1 : D < 100$

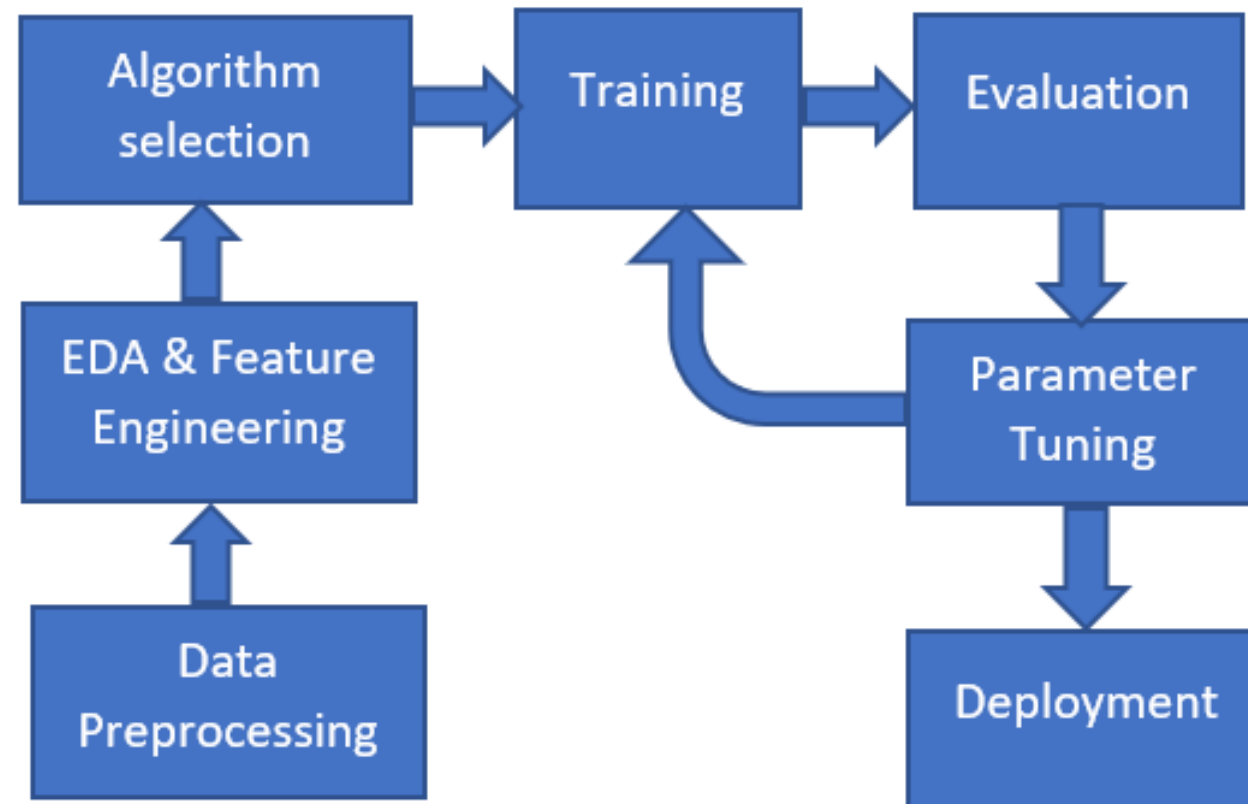
$B2 : 100 \leq D < 120$

$B3 : D \geq 120$

What if I work with a ML algorithm that ONLY takes discrete data ? Put in "buckets".



Depends on the task
(and many other things!!)



Algorithm Selection Example

Training data

15 samples, $\mathbf{s}_i$ for $i = 1..15$

Feature vectors

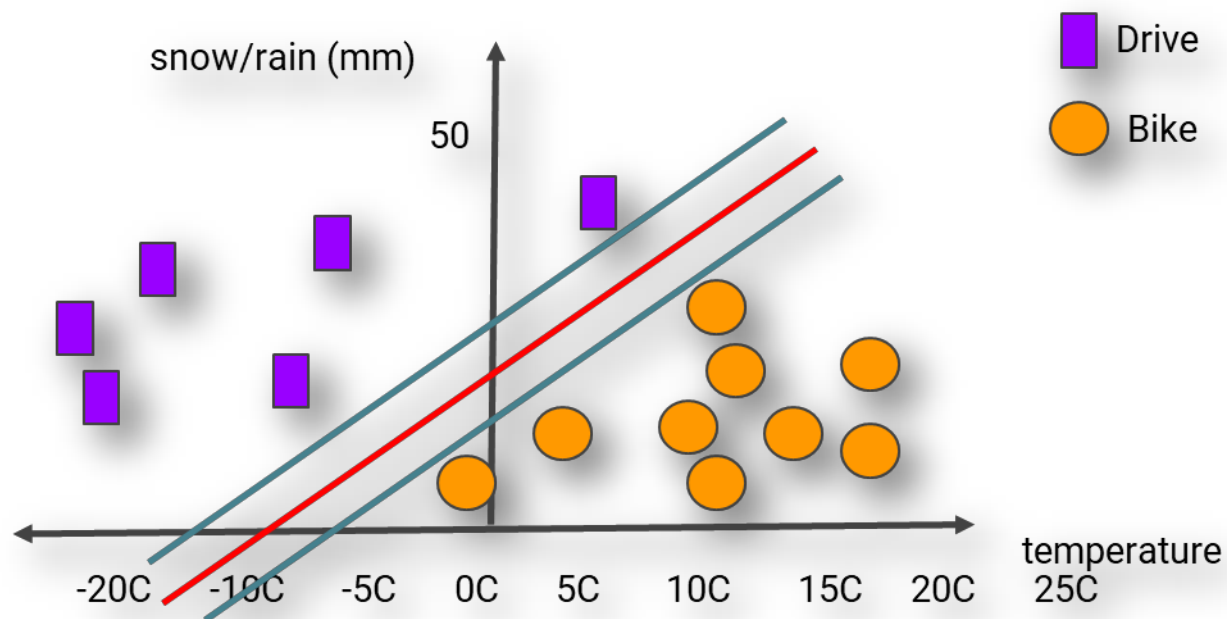
$\vec{\mathcal{X}}$ has 2 dimensions

$\mathcal{X}_1$ Temperature

$\mathcal{X}_2$ Rain/snow

Sample	Temperature	Rain/snow	Bike / Drive
s1	-22	10	Drive
s2	-24	12	Drive
s3	-15	20	Drive
s4	-8	13	Drive
s5	-5	40	Drive
s6	8	45	Drive
s7	-2	5	Bike
s8	5	5	Bike
s9	12	2	Bike
s10	8	5	Bike
s11	12	20	Bike
s12	13	15	Bike
s13	15	10	Bike
s14	18	5	Bike
s15	19	20	Bike

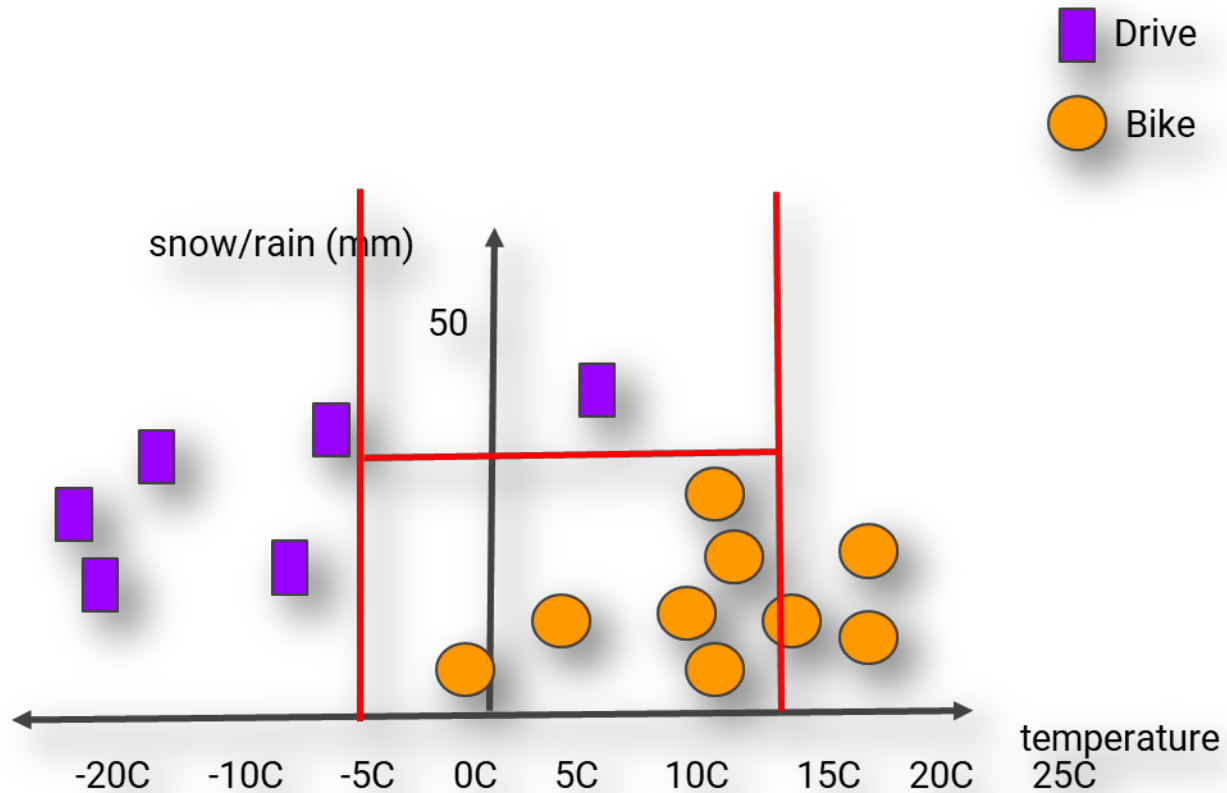
Maximize the margin between the support vectors



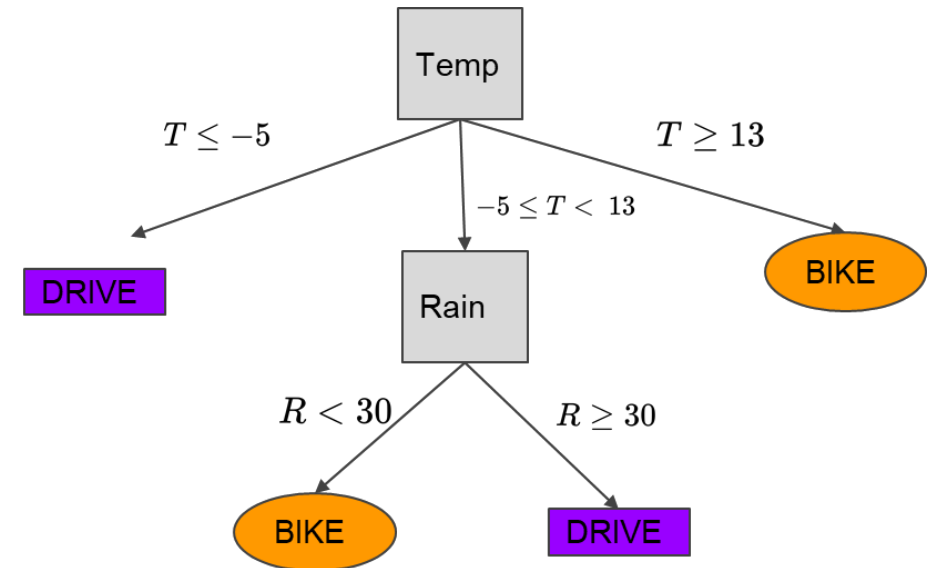
Support Vector Machine

Sensitive to outliers

Will create as many boundaries as necessary to separate the training data.

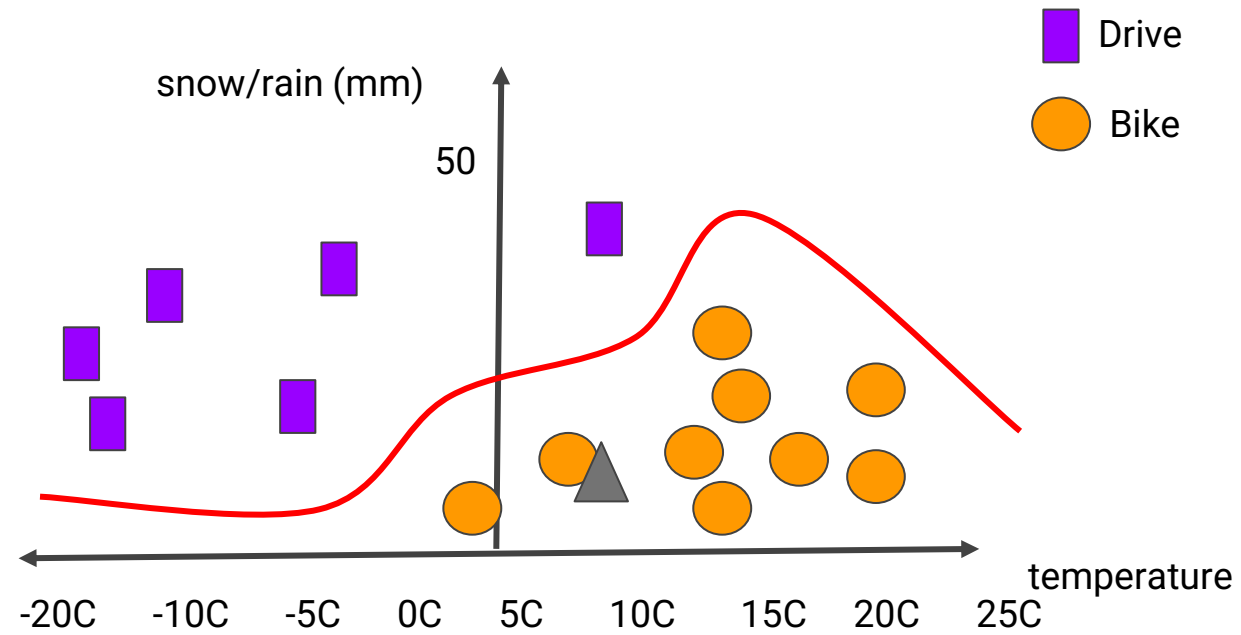


Decision Tree

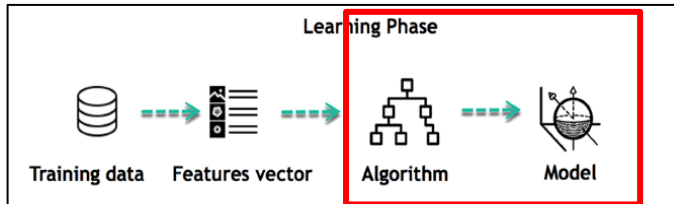


One dimension at a time

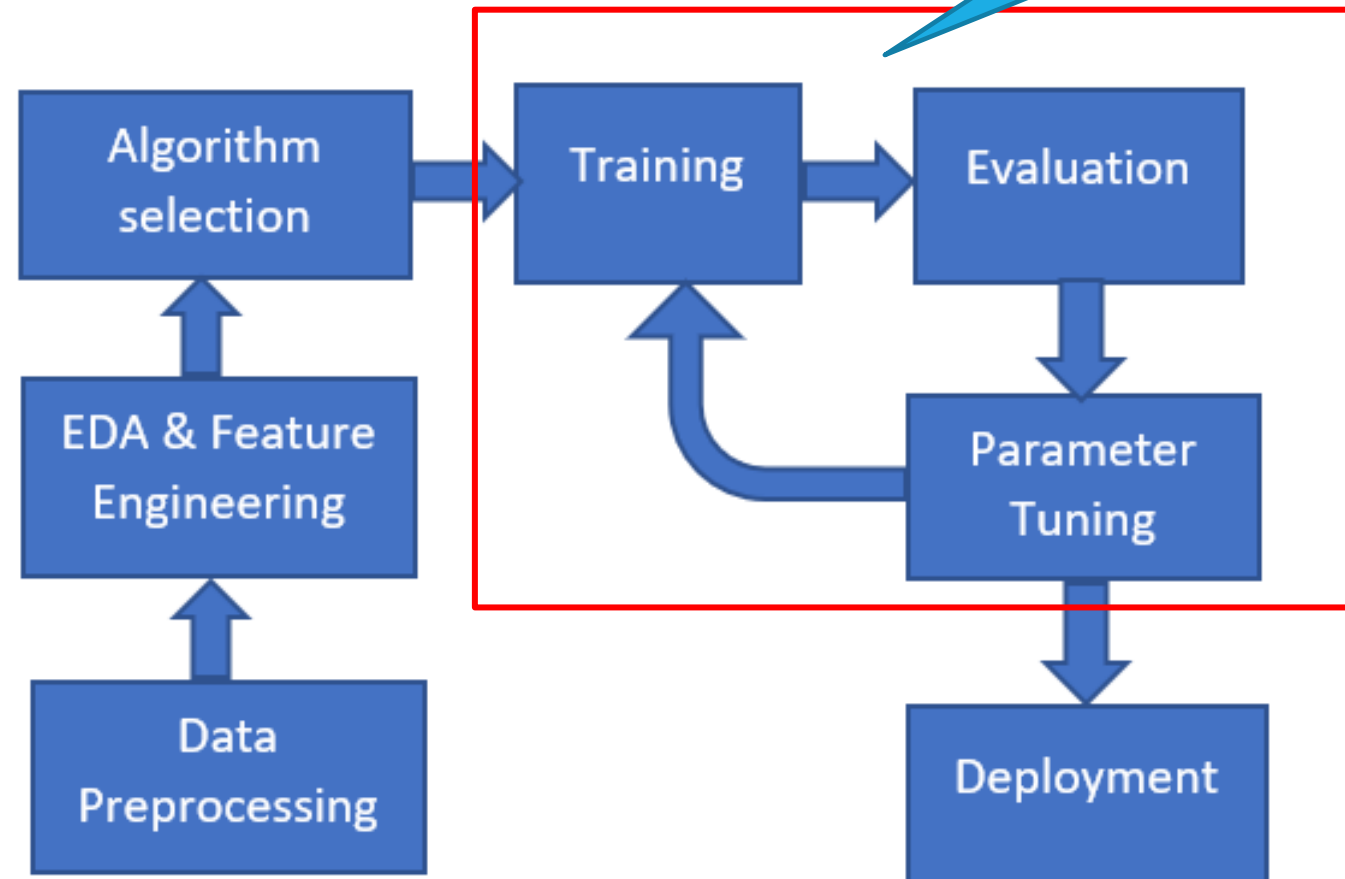
Will be able to learn decision surfaces that are not planes



Multilayer Perceptron



In the next few lectures, we will further look into these steps for different algorithms

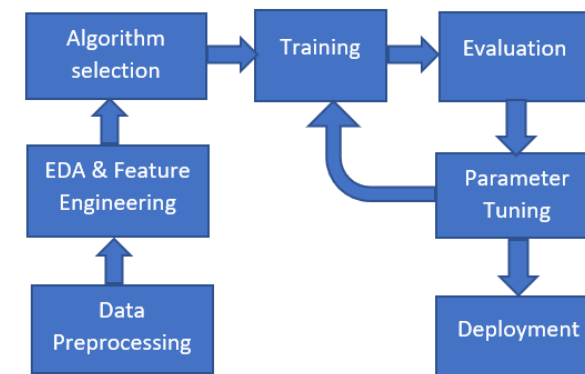


Evaluation

Today is 8C with 2mm of rain.
Tomorrow is 15C with 20mm of rain.

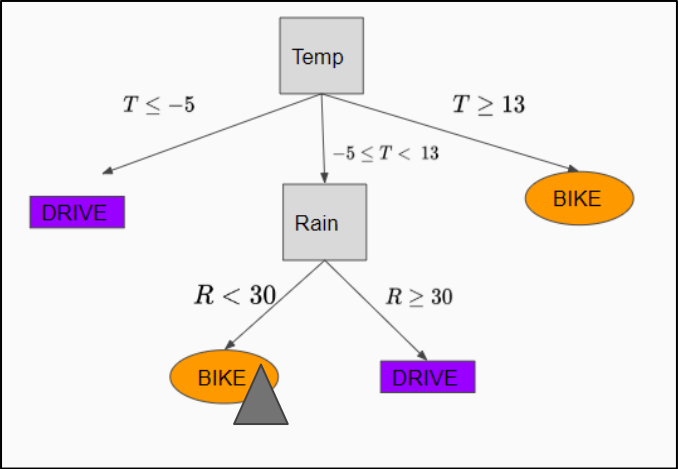
What do the models predict?

Sample	Temperature	Rain/snow	Bike / Drive
T1	8	2	??
T2	15	20	??

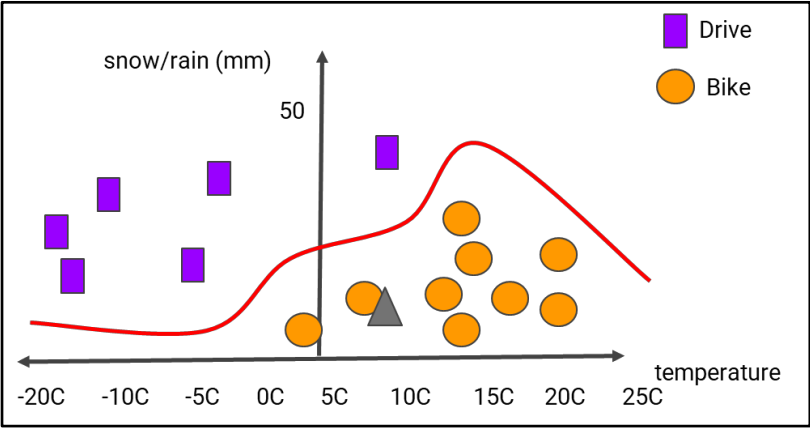


Sample	Temperature	Rain/snow	Bike / Drive
T1 	8	2	??
T2	15	20	??

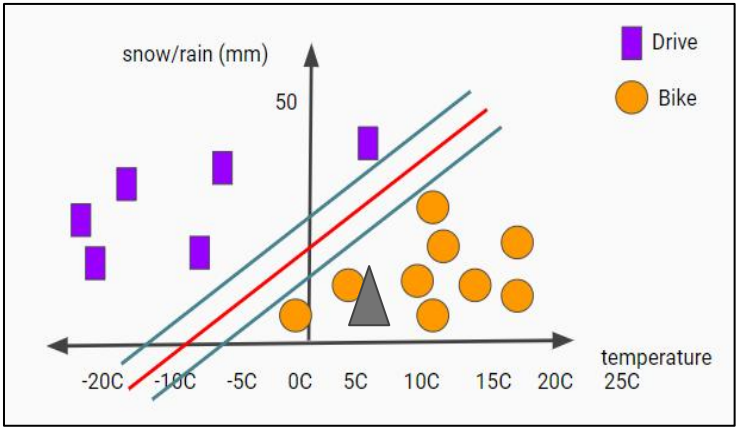
Decision Tree



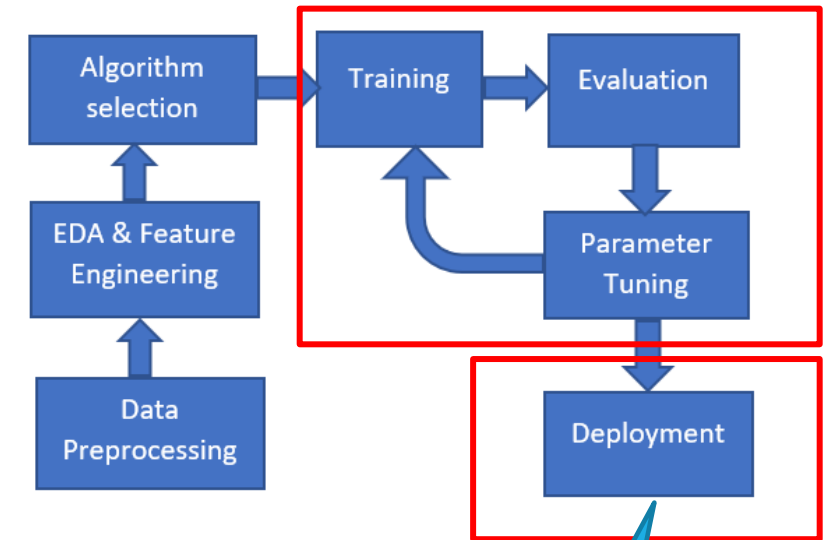
MLP



SVM

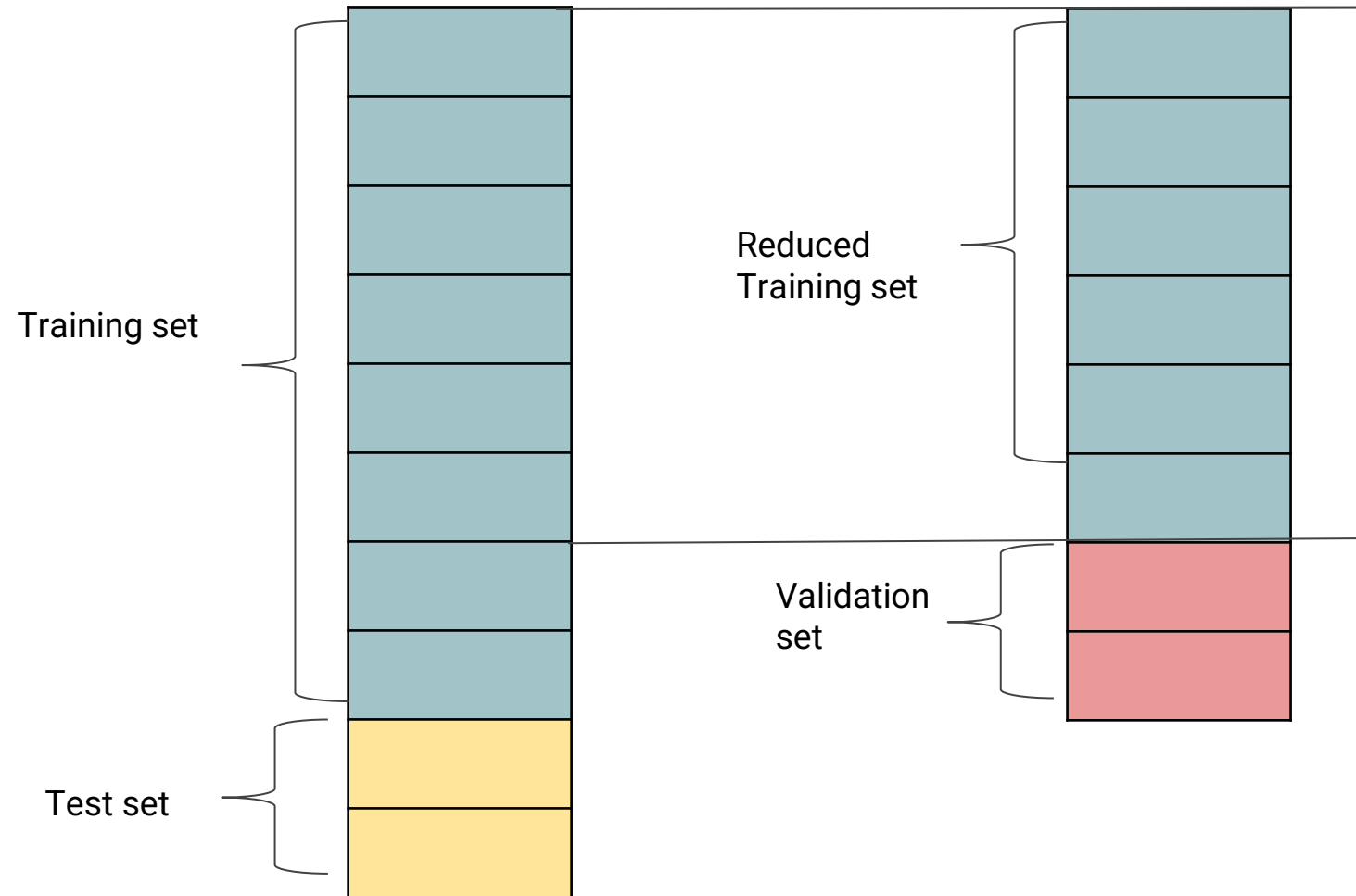


Training set			Prediction	Target
		1	Bike	Drive
		2	Drive	Drive
		3	Drive	Drive
		4	Bike	Drive
		5	Bike	Bike
		6	Drive	Drive
		7	Bike	Bike
Unseen data			Bike	?
			Drive	?



But we should evaluate during training to perform parameter tuning

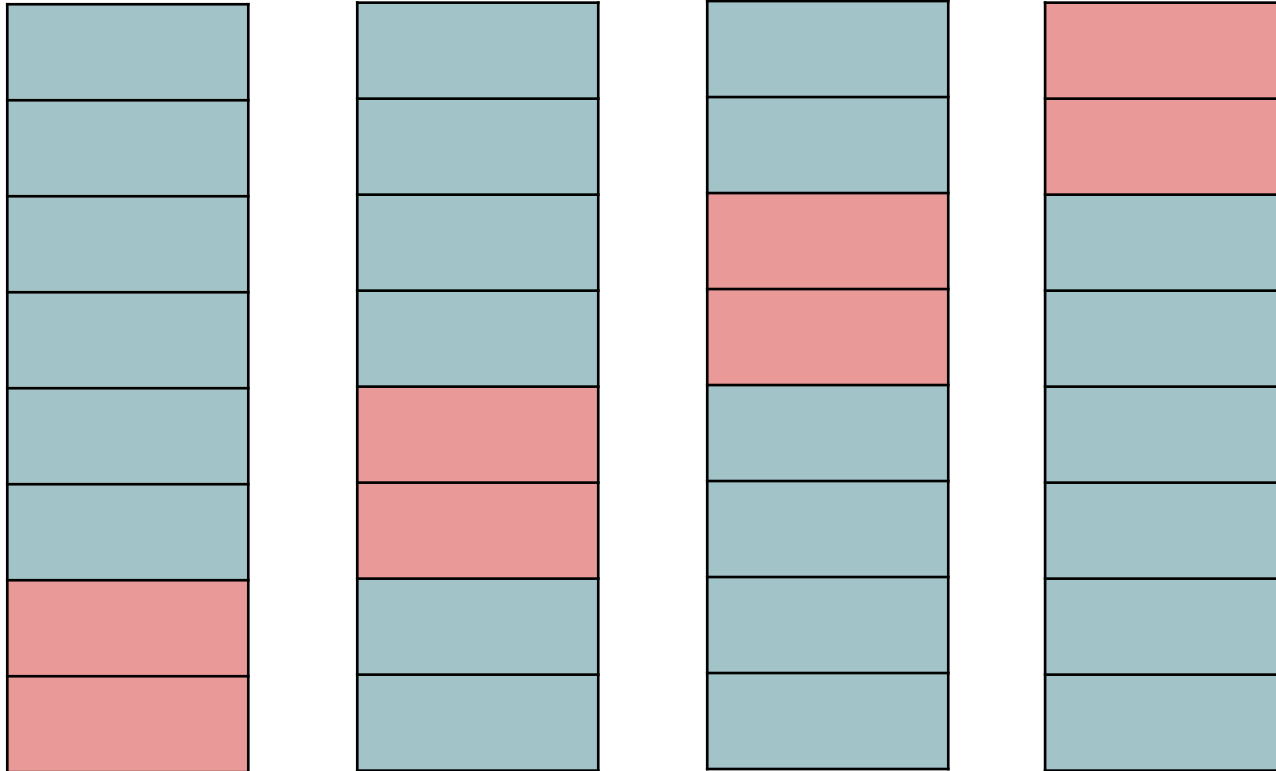
We want good performances here...



Training
set

Validation
set

4-fold cross-validation



Confusion Matrix

		Predicted		
		Bike	Not Bike	
Gold Standard	Bike	Tp = 3	Fn = 3	6
	Not bike	Fp = 1	Tn = 4	5
		4	7	11

Test	Gold Standard	Prediction
1	Bike	Drive
2	Drive	Drive
3	Drive	Drive
4	Bike	Drive
5	Bike	Bike
6	Drive	Drive
7	Bike	Bike
8	Drive	Drive
9	Bike	Drive
10	Bike	Bike
11	Drive	Bike

For classification

$$\text{Precision} = \text{Tp} / (\text{Tp} + \text{Fp})$$

$$= 3 / (3 + 1) = 0.75$$

$$\text{Recall} = \text{Tp} / (\text{Tp} + \text{Fn})$$

$$= 3 / (3 + 3) = 0.5$$

		Predicted		
		Bike	Not Bike	
Gold Standard	Bike	Tp = 3	Fn = 3	6
	Not bike	Fp = 1	Tn = 4	5
		4	7	11

Do the same precision/recall evaluation on the Drive class.

Per class precisions

	Bike	Drive
System	$3/4 = 0.75$	$4/7 = 0.57$

Remember the Evaluation Bias ?

Macro-average : Average on the results per class

Macro-average on precisions: $(\text{Prec}_{C_1} + \text{Prec}_{C_2}) / 2$

$$(0.75 + 0.57) / 2 = 0.66$$

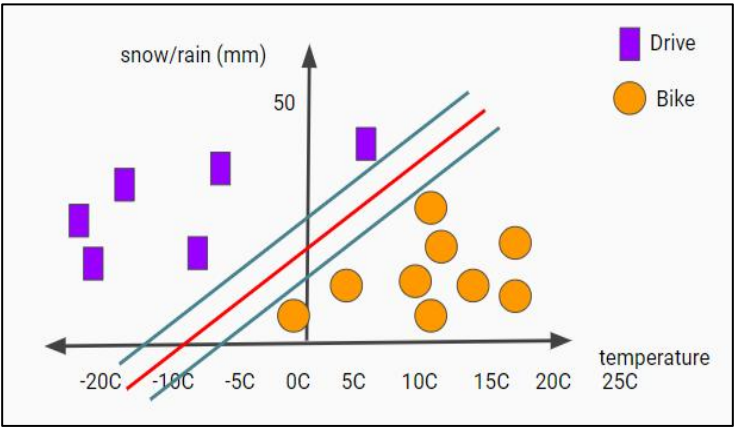
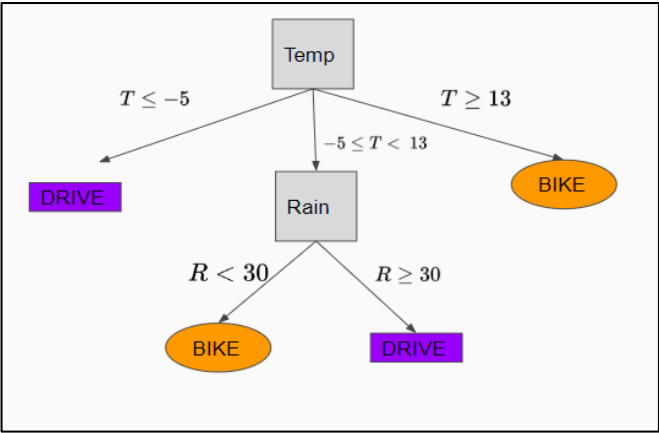
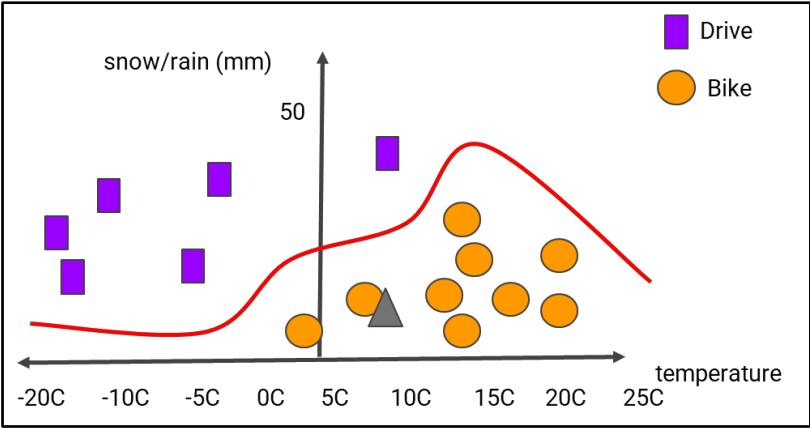
Micro-average : Average when putting all the data together.

Micro-average of precision:: $(\text{TP}_{C_1} + \text{TP}_{C_2}) / (\text{TP}_{C_1} + \text{FP}_{C_1} + \text{TP}_{C_2} + \text{FP}_{C_2})$

$$(3 + 4) / (4 + 7) = 7 / 11 = 0.64$$

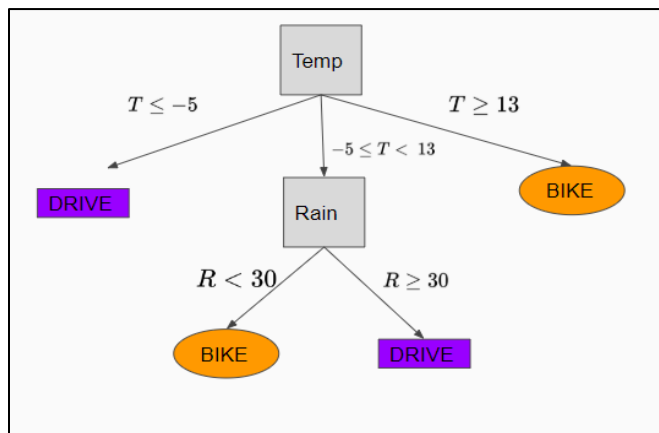
Quiz Question

Which way of combining
more sensitive to class
imbalance



Comparative evaluation

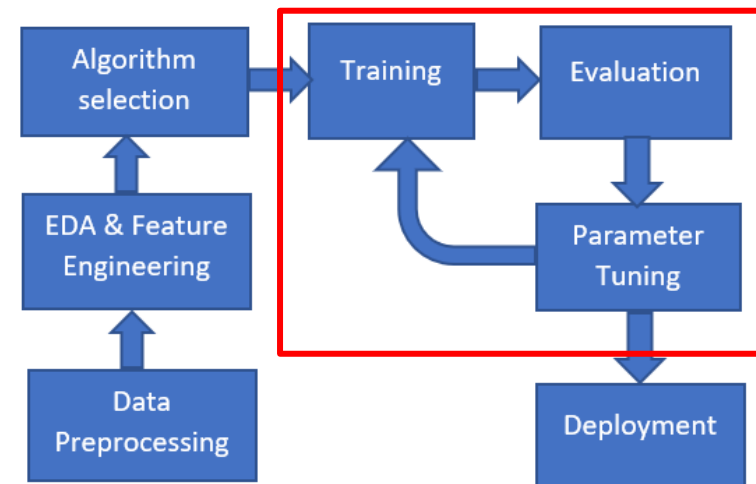
	Bike		Drive	
	Precision	Recall	Precision	Recall
System 1				
System 2				
System 3				



Hyperparameters

Examples:

- Max depth of tree
- Binning or not
- With/Without Outliers
- With/Without aggregate features



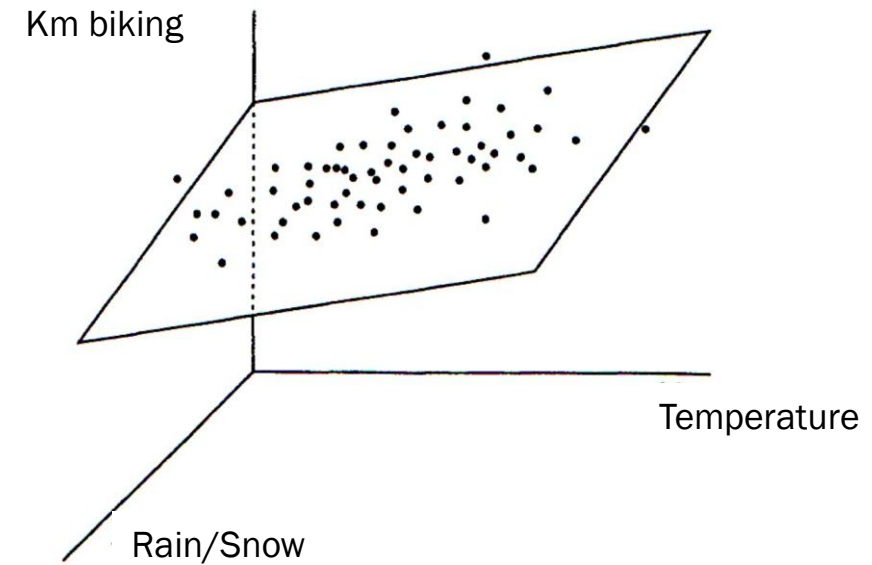
Comparative evaluation for Parameter tuning

	Bike		Drive	
	Precision	Recall	Precision	Recall
System v.1				
System v.2				
System v.3				
....				
System v.n				

Different approaches (e.g. grid search) to explore parameters

Sample	Temperature	Rain/Snow	Km biking
s1	-22	10	2
s2	-24	12	0
s3	-15	20	3
s4	-8	13	4
s5	-5	40	3
s6	8	45	8
s7	-2	5	10
s8	5	5	12
s9	12	2	12
s10	8	5	14
s11	12	20	20
s12	13	15	20
s13	15	10	25
s14	18	5	20
s15	19	20	40

What about evaluation for regression?



Mean Square Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Larger errors are more penalized in MSE than MAE

Mean Absolute Error

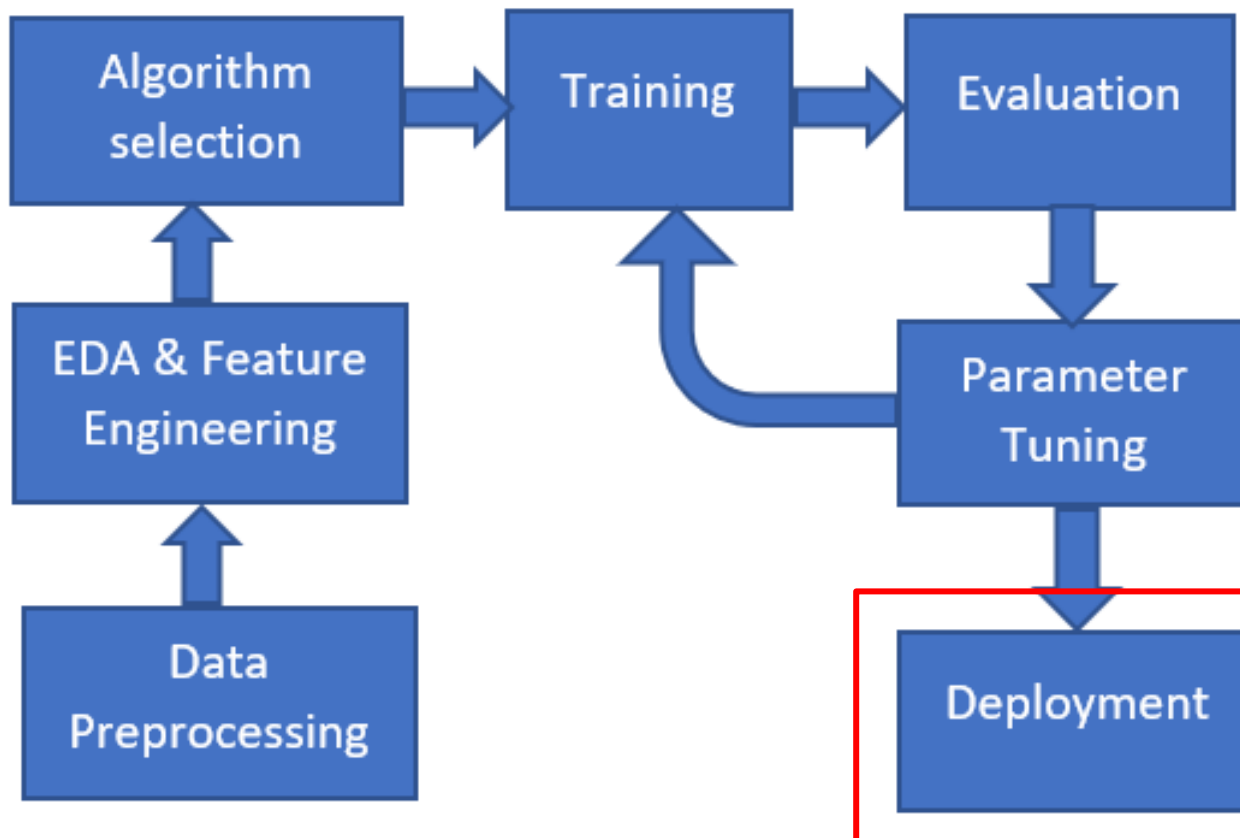
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

R-squared

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Variance in line with original data

Deployment

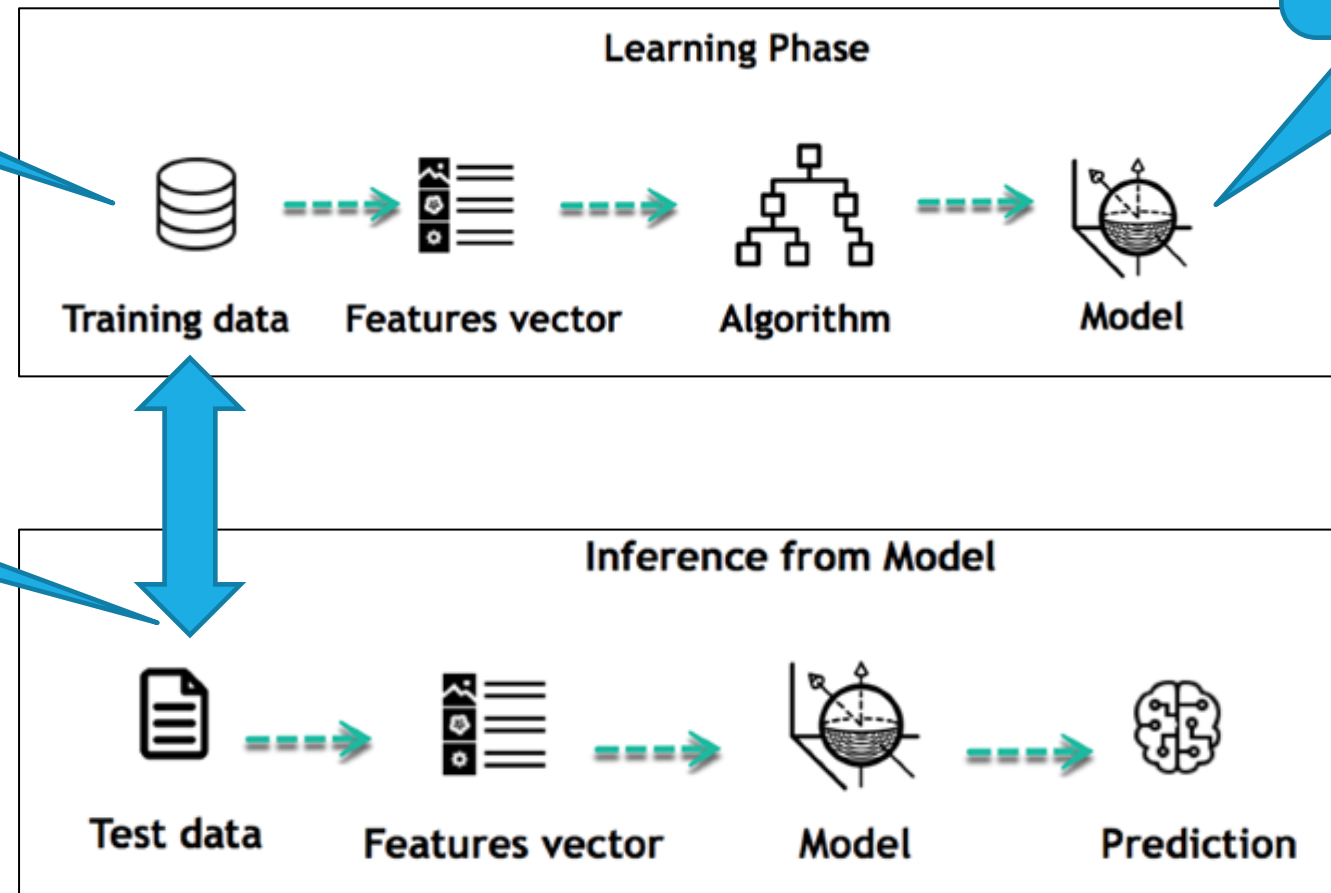


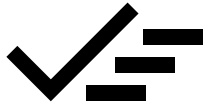
Predictions will need to be performed on unseen data

Training data should be representative of all unseen data

Test data MUST be similar to training data

Model should not be overfitted or underfitted





PREDICTIVE ANALYSIS

- Introduction to Machine Learning
- Supervised Machine Learning steps